

Computer Vision

CO1. Assessment Tool 1 - Short Answer Test.

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Answers:

1. a) Computer vision helps computer understand, identify, analyze, and interpret images and videos automatically using intelligent techniques.
- b) The two real-world applications of computer vision are face recognition system and self-driving cars.
2. a) Feature extraction finds important image details, making computer analysis faster, easier, and more accurate for recognition.
- b) Object recognition helps computer identify object correctly, improving decision-making, automation, and image understanding in applications.
3. a) Low-level image processing improves image quality through noise removal, filtering, enhancement and brightness adjustment techniques.
- b) Mid-level processing extracts features, while high-level processing understands objects, scenes, and image meanings clearly together.
4. a) A pixel is the smallest unit of a digital image containing color and brightness information accurately.
- b) Higher image resolution provides clearer pictures, better details, improved quality, and more accurate image analysis results.

5. a) An image sensor captures light and converts it into digital signals for creating digital images successfully.
- b) Digital cameras and scanners are two common devices used for image acquisition in computer vision systems.
6. a) Sampling converts continuous images into small pixels, creating digital images suitable for computer processing and storage.
- b) Quantization assigns numerical values into pixels, representing image brightness and color information accurately for storage.
7. a) Perspective projection converts 3-d objects into 2-d images while maintaining realistic depth and appearance accurately.
- b) Illumination provides necessary light, affecting image brightness, visibility, quality and object recognition performance significantly always.
8. a) Image processing improves image quality before analysis helping computer vision systems perform tasks more effectively always.
- b) Image processing enhances images, while computer vision understands image content and makes intelligent decisions automatically correctly.
9. a) Computer vision helps doctors, detect diseases by analyzing medical images quickly, accurately and efficiently every day.

b) Computer vision helps autonomous vehicles detect roads, obstacles pedestrians, ensuring safety in navigation and driving in every day.

10. a) Pixels are basic building blocks because every digital image consists of millions of tiny pixels together.

b) Higher sampling and image resolution improve image details, supporting accurate analysis and better recognition performance consistently.